Hands of Heritage: Warli World

Detailed Resume + Portfolio Case Study

An immersive VR learning system for Warli art, cultural context, embodied interaction, and AI-guided personalization.

Focus: Design + Interaction

Overview

- Hands of Heritage: Warli World is an immersive Meta Quest 3 experience that teaches Warli art through a culturally situated VR village, elder-led storytelling, voice interaction, and embodied drawing practice.
- The experience shifts learning from passive observation to active apprenticeship: users enter a Warli village, understand cultural meaning, practice visual grammar, and create a personalized final artwork.
- This version focuses on the main village and drawing experience, excluding the tutorial and envelope sequence.
- The project frames Warli art as a living practice, not just a finished visual artifact, by combining environment, ritual, NPC guidance, and hands-on creation.

Resume Positioning

- Project type: XR learning system, immersive cultural heritage experience, and VR interaction design case study.
- Primary contribution focus: experience design, UI design, interaction flow, AI voice integration, NPC systems, system architecture, and research evaluation framing.
- Best resume framing: not a simple VR drawing app, but a scaffolded cultural-learning system that teaches visual grammar through situated and embodied interaction.
- Core value: users learn by moving, listening, speaking, preparing material, tracing, assembling, and drawing independently.

Problem

- Visual literacy gap: many users can view traditional Indian art but cannot read its symbols, structure, or cultural meaning.
- Static observation problem: videos, museum displays, and textbook references make the art feel distant and finished instead of practiced and lived.
- Heritage erosion problem: intangible cultural practices weaken when younger audiences cannot recognize the visual grammar or ritual context behind the artwork.
- Interaction challenge: the experience needed to teach both cultural context and procedural artistic skill without overwhelming users inside VR.

Design Goal

- Create a VR experience where cultural context, visual grammar, and drawing practice are learned as one connected journey.
- Use the village environment as the interface, not just as decoration.
- Make the elder NPC function like a guide, mentor, and cultural storyteller rather than a generic chatbot.
- Balance beauty, clarity, pacing, and interaction feedback so the user can understand the art before creating it.
- Support a progression from contextual priming to scaffolded practice to freehand creative transfer.

Key Features

- Culturally situated village environment
  - Thatched huts, Warli wall paintings, animals, clay pots, campfire, trees, carts, hay bales, terrain, warm lighting, and spatial audio.
  - Environmental details reinforce that Warli art belongs to village life, ritual, community, and storytelling.
- NPC-guided learning journey
  - A child NPC welcomes the user and leads them toward Ramu, the elder artist.
  - The elder artist teaches Warli through dialogue, visual references, and a three-stop village tour.
- Three-part cultural priming
  - Rice paste stop: introduces material preparation and asks the user about a peaceful place.
  - Shapes and people stop: teaches circle, triangle, square grammar and asks about people, animals, or actions.
  - Village-life art stop: explains motifs such as dance, harvest, animals, and community scenes, then asks about mood.
- Embodied drawing progression
  - Rice grinding ritual, ghost-guided tracing, XR socket-based motif assembly, reference-based drawing, and personalized freehand creation.
  - The flow moves from understanding to applying to creating.
- Personalized final artwork
  - User voice responses are stored and transformed into a personalized Warli drawing prompt.
  - The final task connects cultural form with the user's own memory, background, characters, and mood.

UI Design

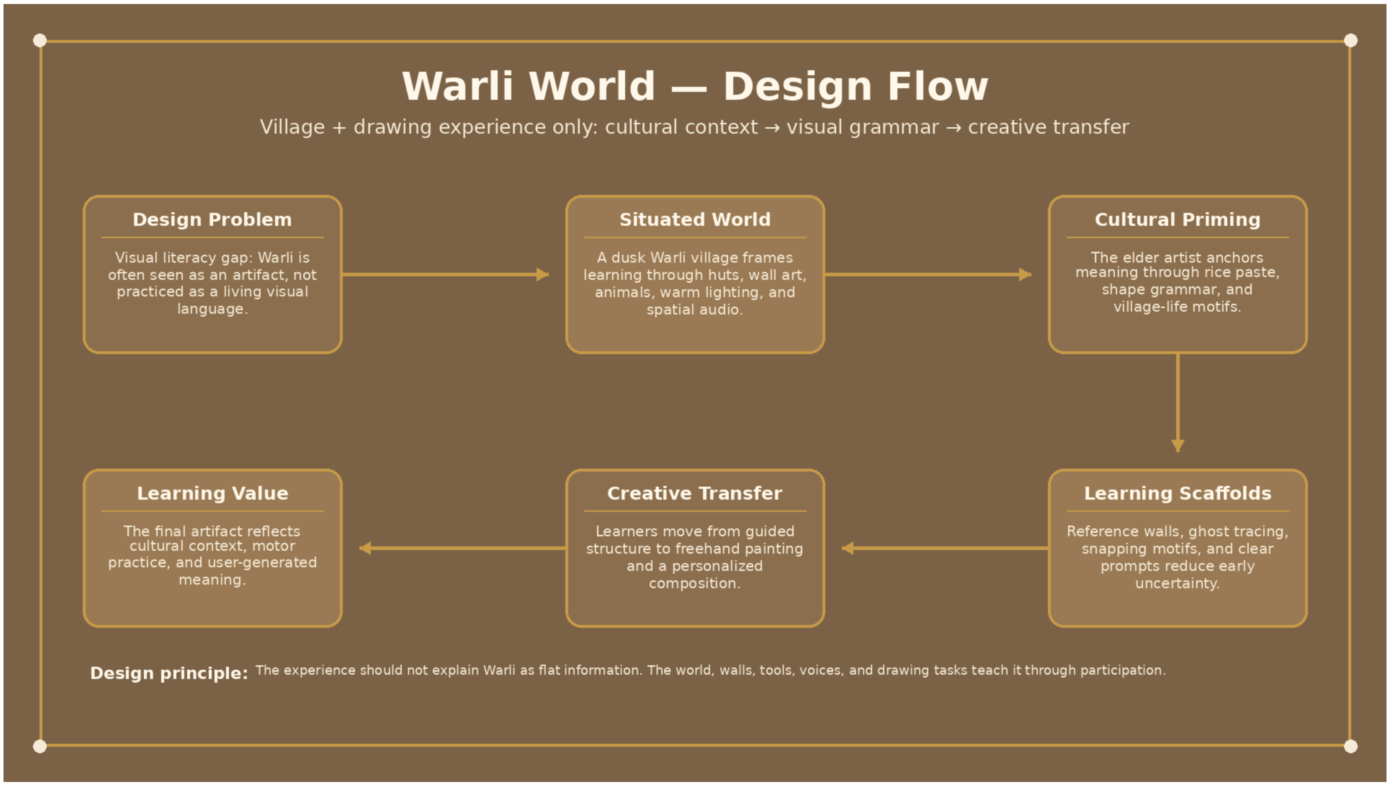
- The UI is designed as in-world guidance rather than a flat application overlay.
- World-space hint panels provide short contextual instructions while preserving immersion.
- Wall-based references turn the village itself into an instructional surface.
- The visual system uses warm brown, cream, and gold tones to match the earthy Warli village aesthetic.
- Subtitles and short prompts support comprehension without forcing the user to read long blocks of text in VR.
- Reference panels and personalized prompt panels guide the transition from learning to independent drawing.
- Submit and clear controls provide user agency during drawing tasks without overcomplicating the interface.
- Screen-space overlay UI was avoided because VR rendering required world-space UI for headset stability.

Design Decisions

- Treat priming as curriculum: the village tour is not a prelude; it is the first learning layer.
- Use environmental references instead of long instruction panels: users learn shape grammar by looking at wall paintings and motif examples.

- Make the elder's teaching conversational: voice dialogue helps the experience feel more like cultural transmission than a software tutorial.
- Keep UI minimal and contextual: the user should know what to do next without feeling like they are inside a menu-heavy app.
- Tie every visual element to learning purpose: huts, walls, materials, and motif examples support cultural context and interaction readiness.

Design Flow Diagram



Design Flow - cultural context, environmental references, elder dialogue, visual grammar, and world-space UI prepare the user for embodied creation.

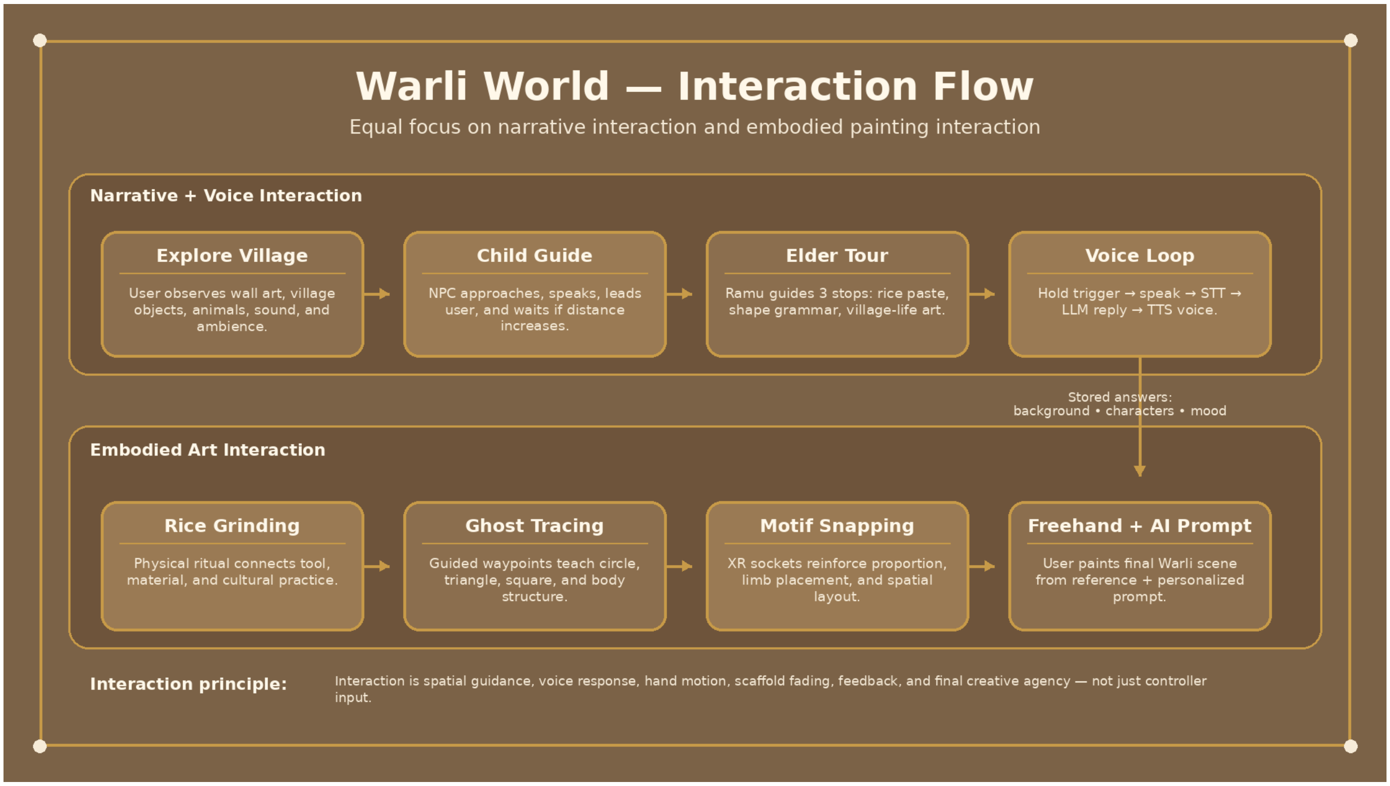
Interaction Design

- The interaction design is built around movement, attention, speech, embodied action, feedback, and creative transfer.
- Users do not only grab objects; they follow NPCs, listen, respond, prepare material, trace geometry, assemble motifs, and draw independently.
- The strongest interaction idea is scaffolded independence: the system gives structure early, then gradually asks the user to create without full guidance.
- Voice interaction makes learning participatory by asking users to reflect on background, characters, activity, and mood.
- Rice grinding creates a physical ritual before painting, making the art process feel material and cultural rather than purely digital.
- Ghost tracing builds geometric confidence and motor memory.
- Motif snapping teaches proportion, placement, and anatomical logic through spatial assembly.
- Freehand drawing tests whether the user can transfer what they learned into an original Warli-style composition.

Interaction Flow

- Village orientation: user looks around, hears ambience, and understands that the environment is part of the learning experience.
- Child guidance: child NPC approaches, welcomes the user, and leads them toward the elder artist.
- Elder tour: Ramu explains materials, visual grammar, and village-life motifs across multiple learning stops.
- Voice response: user speaks answers using trigger-based recording; responses become creative inputs.
- Embodied ritual: user performs rice grinding before painting, linking interaction to cultural preparation.
- Scaffolded practice: user traces shapes and assembles motifs using guided feedback.
- Freehand transfer: user draws from reference, then creates a personalized composition based on their own answers.
- Research logging: interactions such as strokes, timing, coverage, tracing, and task submissions are captured for analysis.

Interaction Flow Diagram



Interaction Flow - user journey from village exploration and voice-based learning to embodied ritual, scaffolded practice, motif assembly, and personalized freehand drawing.

AI Elements

- Speech-to-text: Groq Whisper converts player speech into transcript text.
- LLM dialogue: Groq LLaMA generates short elder NPC responses based on conversation context.
- Text-to-speech: Google Cloud TTS converts child and elder responses into spoken voices.
- Personalization: player answers are stored and used to generate the final Warli drawing prompt.
- AI is used to support conversation and personalization, not to replace the core embodied learning interaction.

Technology Used

- Engine and language: Unity 2022.3 and C#.
- Hardware: Meta Quest 3.
- XR interaction: XR Interaction Toolkit and XR Device Simulator for testing.
- AI and voice: Groq Whisper, Groq LLaMA 3.3, and Google Cloud Text-to-Speech.
- NPCs and animation: Mixamo characters, humanoid rigs, Animator Controllers, walking/talking/pointing states, and waypoint movement.

- Audio: spatial ambience, NPC voice playback, audio fading, and interaction sound effects.
- Research logging: JSON event logs for task timing, brush strokes, trace events, motif assembly, and drawing outputs.
- Version control: Unity Version Control with separate branches for environment, testing, and pre-production.

System Architecture

- Hardware layer: Meta Quest 3 headset and VR controllers.
- XR runtime layer: Unity project configured for VR interaction, scene loading, controller input, and world-space UI.
- Environment layer: Warli village, terrain, huts, painted walls, animals, objects, lighting, and spatial audio zones.
- NPC layer: child guide and elder artist with waypoint movement, animation states, speech playback, and pacing logic.
- Voice layer: microphone recording, WAV conversion, speech-to-text transcription, LLM response generation, and TTS playback.
- Learning layer: cultural priming, rice-paste explanation, shape grammar, village-life motif interpretation, and drawing preparation.
- Interaction layer: rice grinding, brush painting, ghost tracing, socket snapping, freehand canvas drawing, submit, and clear controls.
- Data layer: event logging, phase detection, duration metrics, brush stroke metrics, PNG coverage, survey results, and interview themes.

Voice + Personalization Pipeline

- User holds the VR trigger to start recording speech.
- Unity captures microphone audio and converts the clip into WAV bytes.
- Groq Whisper transcribes the audio into text.
- The transcript is displayed as feedback and stored as part of the user's learning context.
- Groq LLaMA generates a short elder NPC response using the ongoing conversation history.
- Google Cloud TTS converts the response into NPC speech.
- The user's answers about place, characters, activity, and mood are passed into the drawing scene.
- The final prompt asks the user to create a personalized Warli scene based on their own responses.

Case Study + Evaluation

- The evaluation compared two learning orders: Scaffolding -> Freehand and Freehand -> Scaffolding.
- The study examined whether guided practice improved later freehand fluency, efficiency, and visual fidelity.
- Logged metrics included task duration, brush strokes, canvas coverage, tracing events, motif assembly events, and final drawing outputs.

- The analysis pipeline grouped events by task, detected Baseline, Test 1, and Test 2 phases, and computed learning deltas across phases.
- Survey and interview data were used to understand perceived effort, confidence, cultural connection, and interaction friction.
- The safest resume framing is directional: scaffold-first users showed faster later task completion, but the overall comparison should not be framed as statistically significant unless confirmed in the final report.

Case Study Data Points

- Mean Duration Test 1: Freehand -> Scaffolding = 217.53 sec; Scaffolding -> Freehand = 193.32 sec.
- Mean Duration Test 2: Freehand -> Scaffolding = 195.92 sec; Scaffolding -> Freehand = 152.59 sec.
- Mean Brush Strokes Test 1: Freehand -> Scaffolding = 655.17; Scaffolding -> Freehand = 805.83.
- Mean Brush Strokes Test 2: Freehand -> Scaffolding = 779.50; Scaffolding -> Freehand = 633.67.
- Mean PNG Coverage Test 1: Freehand -> Scaffolding = 0.0252; Scaffolding -> Freehand = 0.0270.
- Mean PNG Coverage Test 2: Freehand -> Scaffolding = 0.0275; Scaffolding -> Freehand = 0.0216.
- Scaffolding -> Freehand users completed Test 2 about 22 percent faster than Freehand -> Scaffolding users.
- The overall t-test did not show a statistically significant difference across the full participant pool, so this should be described as a trend.

Qualitative Findings

- Priming built early mental models: users understood circle, triangle, square logic before the drawing task began.
- The rice grinding ritual helped users feel that painting was a cultural process, not only a digital task.
- Scaffolding supported learning transfer by helping users recall body proportions and motif structure during freehand drawing.
- NPC pacing affected attention and flow; users needed more time to follow, listen, and respond.
- Brush instability and canvas-depth ambiguity consumed cognitive bandwidth and affected drawing confidence.
- Users preferred active making over passive observation because the experience helped them feel connected to the art form.

Key Design Challenges

- Creating cultural context without turning the experience into a lecture.
- Balancing visual beauty with instructional clarity inside the village environment.
- Making Warli motifs understandable for users with no prior exposure.
- Preventing UI from feeling like a flat app overlay inside VR.
- Keeping prompts short enough for headset comfort while still guiding the user clearly.

- Using the village as a learning interface through wall paintings, objects, materials, and NPC dialogue.
- Designing a consistent visual language using earthy colors, warm light, and Warli-inspired references.

Key Interaction Challenges

- NPC walking speed and dialogue pacing could outpace user attention during village exploration.
- Object placement and collisions affected movement and sometimes made users feel stuck.
- Brush instability caused rotation issues, dropping, and unpredictable drawing control.
- Canvas depth ambiguity made it difficult for users to judge when the brush was actually touching the wall.
- Voice echo occurred when the microphone captured NPC text-to-speech audio.
- Animation root motion created position drift, requiring character position locking.
- Straight-line NPC movement caused wall clipping, requiring manually placed waypoint routes.
- Screen-space UI caused VR rendering problems, requiring world-space UI attached to the XR camera.

Design Improvements Identified

- Add distance-aware NPC pacing so the elder waits when the user is behind or distracted.
- Add optional repeat and pause controls for dialogue without forcing users through long explanations.
- Stabilize the brush by anchoring rotation more firmly to the hand/controller.
- Add a visual canvas-distance indicator to reduce depth ambiguity during painting.
- Reduce verbose instructions and replace them with more environmental cues and short prompts.
- Expand wall-painting examples to teach more Warli motifs and visual patterns.
- Branch the experience based on whether the user already knows Warli art or is completely new to it.
- Add adaptive scaffolding that fades based on user confidence and stroke quality.

Detailed Resume Entry

- Hands of Heritage: Warli World | Unity, C#, Meta Quest 3, XR Interaction Toolkit, Groq Whisper/LLaMA, Google Cloud TTS.
- Designed and developed an immersive VR learning system for Warli art, combining a culturally situated village environment, elder-led storytelling, AI-driven voice interaction, and embodied drawing practice.
- Built the experience around contextual priming, visual grammar instruction, rice-paste ritual, ghost-guided tracing, XR socket-based motif assembly, and personalized freehand creation.
- Integrated a voice AI pipeline using Groq Whisper for speech-to-text, Groq LLaMA for elder NPC dialogue, and Google Cloud TTS for spoken responses.
- Designed world-space UI, environmental reference systems, NPC waypoint behavior, voice-based personalization, and scene architecture to support learning flow across exploration, instruction, practice, and creation.

- Evaluated the experience through a counterbalanced user study comparing Scaffolding -> Freehand and Freehand -> Scaffolding learning orders, with logged metrics for duration, brush strokes, coverage, motif events, and learning-transfer analysis.

Resume Bullet Options

- Designed a Meta Quest 3 VR learning experience for Warli art, creating a culturally situated village environment with elder-led storytelling, spatial audio, world-space UI, and environmental reference art to move users from observation to participation.
- Built AI-driven NPC interaction using Groq Whisper, Groq LLaMA, and Google Cloud TTS, enabling users to speak with an elder artist, answer reflective prompts, and generate personalized Warli drawing tasks from their own responses.
- Designed the core learning flow across contextual priming, rice-paste ritual, shape grammar instruction, ghost-guided tracing, XR socket-based motif assembly, and freehand drawing to support progression from guided practice to independent creative transfer.
- Implemented NPC and scene systems in Unity/C#, including waypoint-based movement, talking/walking/pointing animation states, position locking, scene transitions, world-space hint UI, and VR-safe feedback systems for Meta Quest 3.
- Evaluated the experience through a counterbalanced user study comparing Scaffolding -> Freehand and Freehand -> Scaffolding learning orders, logging duration, brush strokes, canvas coverage, motif events, and survey/interview data.
- Identified and resolved major XR design and interaction challenges including screen-space UI crashes, NPC pacing, animation root-motion drift, voice echo, waypoint navigation, brush instability, canvas-depth ambiguity, and cognitive overload from verbose instructions.

One-Line Portfolio Summary

- Hands of Heritage: Warli World is an immersive VR learning system that teaches Warli art through a culturally situated village, elder-led AI voice dialogue, embodied material ritual, scaffolded drawing interactions, and personalized freehand creation.